

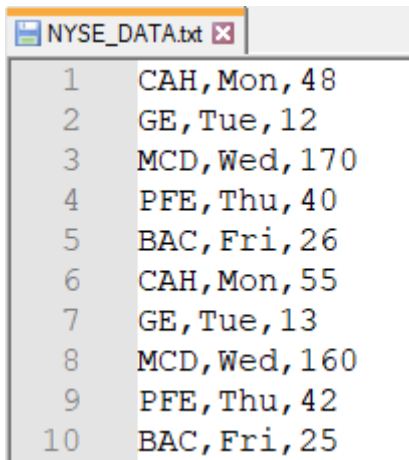
MapReduce Algorithm – Part 1

New York Stock Exchange (NYSE) Trade Data



NYSE Trade Data

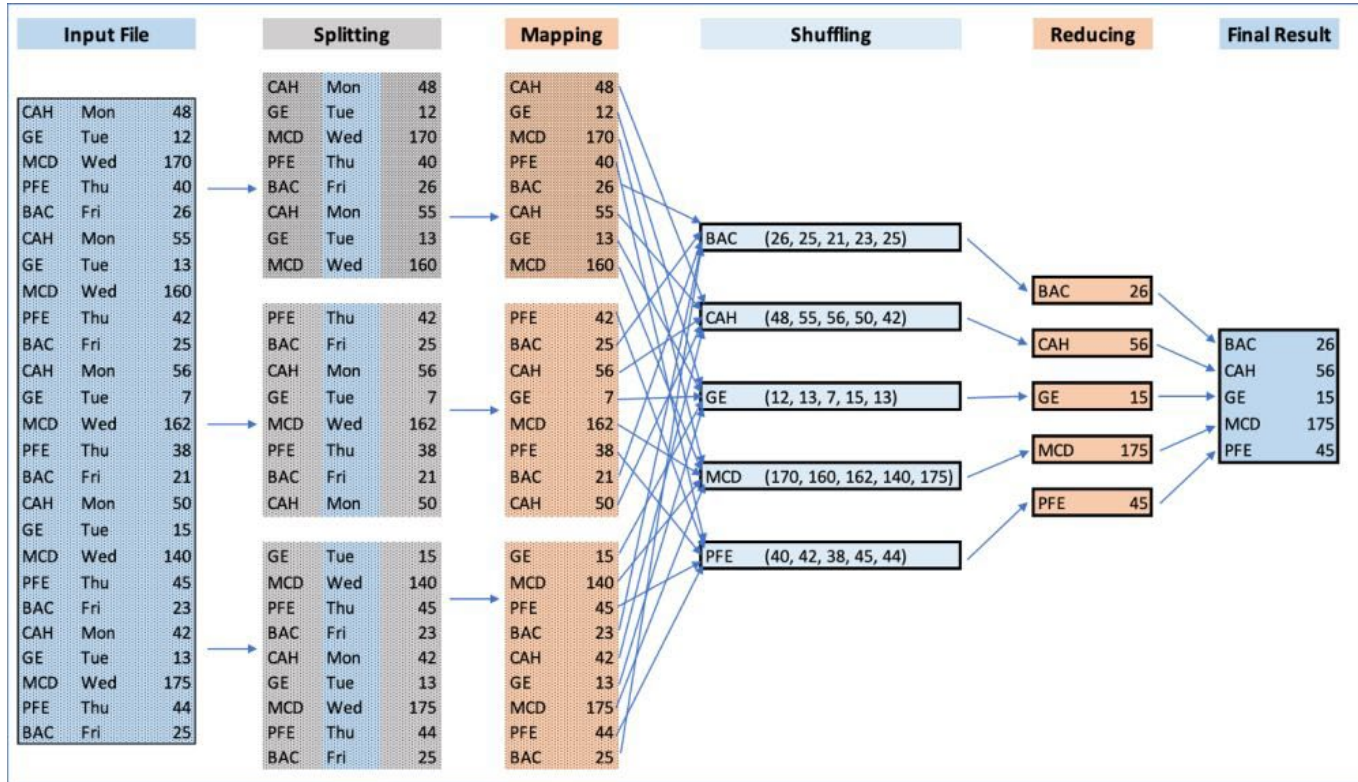
Assume you have five stocks and their trading values at the end of each day during a given week. You want to find the maximum value of the week for each stock.



1	CAH, Mon, 48
2	GE, Tue, 12
3	MCD, Wed, 170
4	PFE, Thu, 40
5	BAC, Fri, 26
6	CAH, Mon, 55
7	GE, Tue, 13
8	MCD, Wed, 160
9	PFE, Thu, 42
10	BAC, Fri, 25



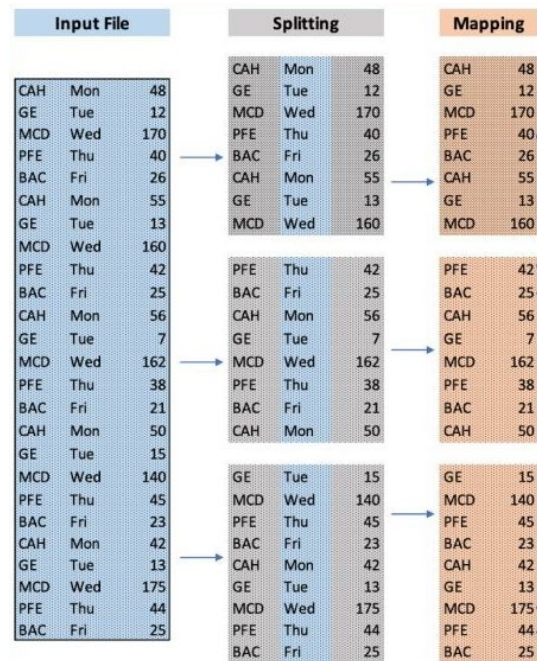
Map Reduce Process – NYSE Trade Data



Map Reduce Process – Mapper Stage

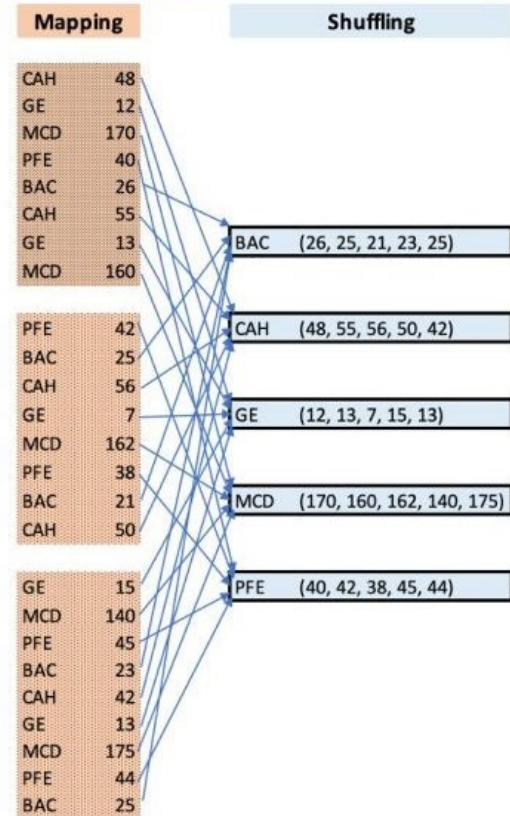
```
5     def mapper(self, _, line):
6         (ticker, day, tradePrice) = line.split(',')
7         yield ticker, int(tradePrice)
```

```
NYSE.py x
1  from mrjob.job import MRJob
2
3  class NYSE(MRJob):
4
5      def mapper(self, _, line):
6          (ticker, day, tradePrice) = line.split(',')
7          yield ticker, int(tradePrice)
8
9      def reducer(self, ticker, Prices):
10         yield ticker, max(Prices)
11
12  if __name__ == '__main__':
13      NYSE.run()
14
15  #!python NYSE.py NYSE_DATA.txt
16  #python NYSE.py NYSE_DATA.txt
```



Map Reduce Process – Sort and Shuffle Stage

```
NYSE.py x
1 from mrjob.job import MRJob
2
3 class NYSE(MRJob):
4
5     def mapper(self, _, line):
6         (ticker, day, tradePrice) = line.split(',')
7         yield ticker, int(tradePrice)
8
9     def reducer(self, ticker, Prices):
10        yield ticker, max(Prices)
11
12 if __name__ == '__main__':
13     NYSE.run()
14
15 #!python NYSE.py NYSE_DATA.txt
16 #python NYSE.py NYSE_DATA.txt
```



Map Reduce Process – Reducer Stage

```
9      def reducer(self, ticker, Prices):  
10         yield ticker, max(Prices)
```

```
NYSE.py x  
1  from mrjob.job import MRJob  
2  
3  class NYSE(MRJob):  
4  
5      def mapper(self, _, line):  
6          (ticker, day, tradePrice) = line.split(',')  
7          yield ticker, int(tradePrice)  
8  
9      def reducer(self, ticker, Prices):  
10         yield ticker, max(Prices)  
11  
12  if __name__ == '__main__':  
13      NYSE.run()  
14  
15  #!python NYSE.py NYSE_DATA.txt  
16  #python NYSE.py NYSE_DATA.txt
```

